

Introduction To MYSQL

SQL is built by 'Database'. To communicate the Database by "MySQL". Oracle, etc., "MySQL" is the base language. MySQL is stored in Structured of "Rows" and "Columns" like a Table.

General Database Structure

Level	Description	Example
Database software	Application used to install, create and manage databases and to run SQL commands.	MySQL, PostgreSQL, MS SQL Server, Oracle
Database	A named container that keeps related data organised and separate from other data.	Movies_db
Schema	The logical blueprint of a database: its tables, columns, datatypes and relationships.	movies(rel_year, mov_name, act_name, lang, dir_name)
Tables	The structure that stores data in rows (records) and columns (fields).	movies table
Data	The actual values stored in the table cells such as names, ages, prices and dates.	1960, 'Kalathur Kannamma', 'Kamal Haasan'

Order: Database software → Database → Schema → Tables → Data

There are different types of MySQL:

DDL (Data definition language)

Purpose: Defines, modifies, and alters the actual structure or schema of the database.

Core Commands:

- CREATE (makes new tables)
- ALTER (modifies table structures)
- DROP (deletes an entire table permanently)
- TRUNCATE (removes all records but keeps the structure)

DML (Data Manipulation language)

Purpose: Manages and modifies the actual data rows stored inside the existing tables.

Core Commands:

- INSERT (adds new rows)
- UPDATE (edits existing data)
- DELETE (removes specific records).

TCL (Transaction Control Language)

Purpose: Manages transactions to group multiple tasks together, ensuring database integrity.

Core Commands:

- COMMIT (saves changes permanently)
- ROLLBACK (undoes changes if an error occurs)

Datatypes of SQL:

NUMERIC	DATE AND TIME	STRING
int	date and time("Y:M:D")	varchar
bigint	timestamp (change in universal time)	char
float		

Movie Database Management and Analytical Reporting Case Study:

#Create a database and use the database

```
create database Movies_db;
use Movies_db;

create table movies
(
  rel_year int,
  mov_name varchar(100),
  act_name varchar(50),
  lang varchar(30),
  dir_name varchar(100)
);
```

#Insert the data's

```
INSERT INTO MOVIES VALUES (1960,'Kalathur Kannamma','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES VALUES (1962,'Kannum Karolum','Kamal Haasan','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES VALUES (1962,'Paadha Kaanikkai','Kamal Haasan','Tamil','K. Shankar');
INSERT INTO MOVIES VALUES (1962,'Paarthaal Pasi Theerum','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES VALUES (1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy');
INSERT INTO MOVIES VALUES (1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan');
INSERT INTO MOVIES VALUES (1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam');
INSERT INTO MOVIES VALUES (1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan');
INSERT INTO MOVIES VALUES (1971,'Anubhavgal Paalichakal','Mammootty','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES VALUES (1972,'Kurathi Magan','Kamal Haasan','Tamil','K. S. Gopalakrishnan');
INSERT INTO MOVIES VALUES (1973,'Arangetram','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES VALUES (1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan');
INSERT INTO MOVIES VALUES (1973,'Sollathaana Ninaikkiren','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES VALUES (1974,'Anbu Thangai','Kamal Haasan','Tamil','S. P. Muthuraman');
INSERT INTO MOVIES VALUES (1974,'Aval Oru Thodar Kathai','Kamal Haasan','Tamil','K. Balachander');
```

1. Retrieve all details from the movie table.

```
select * from MOVIES;
```

Solution:

rel_year	mov_name	act_name	lang	dir_name
1960	Kalathur Kannamma	Kamal Haasan	Tamil	A. Bhimsingh
1962	Kannum Karalum	Kamal Haasan	Malayalam	K. S. Sethumadhavan
1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
1962	Paarthaal Pasi Theerum	Kamal Haasan	Tamil	A. Bhimsingh
1963	Anandha Jodhi	Kamal Haasan	Tamil	V. N. Reddy
1963	Vanambadi	Kamal Haasan	Tamil	G. R. Nathan
1970	Maanavan	Kamal Haasan	Tamil	M. A. Thirumugam
1971	Annai Velankanni	Kamal Haasan	Tamil	Thankappan
1971	Anubhavangal Paalichakal	Mammootty	Malayalam	K. S. Sethumadhavan
1972	Kurathi Magan	Kamal Haasan	Tamil	K. S. Gopalakrishnan
1973	Arangetram	Kamal Haasan	Tamil	K. Balachander
1973	Kaalachakram	Mammootty	Malayalam	K. Narayanan
1973	Sollathaana Ninaikkiren	Kamal Haasan	Tamil	K. Balachander
1974	Anbu Thangai	Kamal Haasan	Tamil	S. P. Muthuraman
1974	Aval Oru Thodar Kathai	Kamal Haasan	Tamil	K. Balachander

2. Display only movie names and actor names.

```
SELECT mov_name, act_name  
FROM movies;
```

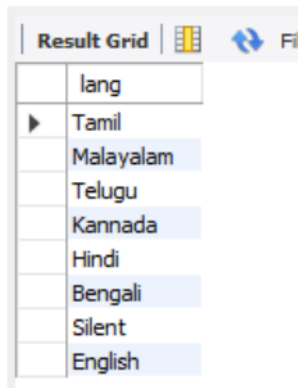
Solution:

	mov_name	act_name
►	Kalathur Kannamma	Kamal Haasan
	Kannum Karalum	Kamal Haasan
	Paadha Kaanikkai	Kamal Haasan
	Paarthaal Pasi Theerum	Kamal Haasan
	Anandha Jodhi	Kamal Haasan
	Vanambadi	Kamal Haasan
	Maanavan	Kamal Haasan
	Annai Velankanni	Kamal Haasan
	Anubhavangal Paalichakal	Mammootty
	Kurathi Magan	Kamal Haasan
	Arangetram	Kamal Haasan
	Kaalachakram	Mammootty
	Sollathaana Ninaikkiren	Kamal Haasan
	Anbu Thangai	Kamal Haasan
	Aval Oru Thodar Kathai	Kamal Haasan
	Gumasthavin Magal	Kamal Haasan
	Kanyakumari	Kamal Haasan

3. Display distinct movie languages.

```
select distinct lang from movies;
```

Solution:



The screenshot shows a 'Result Grid' window with a table containing the results of the query. The table has one column labeled 'lang' and ten rows of language names. The first row is 'Tamil', followed by 'Malayalam', 'Telugu', 'Kannada', 'Hindi', 'Bengali', 'Silent', and 'English'. The last two rows are empty.

lang
Tamil
Malayalam
Telugu
Kannada
Hindi
Bengali
Silent
English

4. Retrieve unique actor names.

```
SELECT DISTINCT act_name  
FROM movies;
```

Solution:



The screenshot shows a 'Result Grid' window with a table containing the results of the query. The table has one column labeled 'act_name' and twenty rows of actor names. The names are: Kamal Haasan, Mammootty, Rajinikanth, Mohanlal, Vijay, Silambarasan, Ajith Kumar, Suriya, Trisha, Nayanthara, Karthi, Dhanush, Dulquer Salm..., Amala Paul, Rashmika Ma..., Manju Warior, Sai Pallavi, Parvathy Thi..., and Vijay Sethup... The last two rows are empty.

act_name
Kamal Haasan
Mammootty
Rajinikanth
Mohanlal
Vijay
Silambarasan
Ajith Kumar
Suriya
Trisha
Nayanthara
Karthi
Dhanush
Dulquer Salm...
Amala Paul
Rashmika Ma...
Manju Warior
Sai Pallavi
Parvathy Thi...
Vijay Sethup...